

Week 3

Old architecture based on LorentzNet

x = Cartesian particle 4-momenta, ϕ = learnable functions, $\psi(\cdot) = \text{sgn}(\cdot) \log(|\cdot| + 1)$,
 c = hyperparameter, $N = \#$ particles, $\|\cdot\|, \langle \cdot \rangle$ = Minkowski norm and inner product

LGBE ([1])

$$m_{ij}^l = \phi_e(h_i^l, h_j^l, \psi(\|x_i^l - x_j^l\|), \psi(\langle x_i^l, x_j^l \rangle))$$

$$m_{ij}^l = \phi_m(m_{ij}^l) \cdot m_{ij}^l$$

$$x_i^{l+1} = x_i^l + \frac{c}{N} \sum_{j \in [N]} \phi_x(m_{ij}^l) \cdot |x_i - x_j|$$

$$h_i^{l+1} = h_i^l + \phi_h(h_i^l, \sum_{j \in [N]} \phi_w(m_{ij}^l) m_{ij}^l)$$

Simple LE block this week

$$m_{ij}^l = \phi_e(\psi(\|x_i^l - x_j^l\|), \psi(\langle x_i^l, x_j^l \rangle))$$

$$m_{ij}^l = \phi_m(m_{ij}^l) \cdot m_{ij}^l$$

$$x_i^{l+1} = x_i^l + \frac{c}{N} \sum_{j \in [N]} \phi_x(m_{ij}, g) \cdot |x_j^l - x_i^l|$$

g = global feature vector (currently time embedding $\phi_t(t)$)

New architecture I

1. Embed scalar time value to random Fourier features through two fully connected layers with 32 and 64 nodes (following [2])
2. Concatenate time embedding with jet conditioning information, which consists of:
 - 2.1 One-hot encoded jet type
 - 2.2 Jet mass
 - 2.3 Jet transverse momentum
 - 2.4 Jet number of constituents
3. Pass this through a fully connected layer of size 64 to produce the initial global embeddings g_0
4. Concatenate g_0 with features $x_1^0 \dots x_N^0$ initialized from the initial distribution p_0 .
5. Now, we use 6 Lorentz-group equivariant layers.
 - 5.1 The input to the $l + 1$ -th layer is g^l and $x_n^l \dots x_N^l$.
 - 5.2 Calculate messages $m_{ij}^l = \phi_m(\psi(\|x_i^l - x_j^l\|), \psi(\langle x_i^l, x_j^l \rangle))$

New architecture II

5.3 Update the global embedding $g_{l+1} = \phi_{eg}(g_l, \frac{c}{N} \sum_{i,j \in [N]} \phi_e(m_{ij}) \oplus \sum_{i,j \in [N]} \phi_e(m_{ij}))$

5.4 Feed in the updated global embedding as well as the original embedding to update the graph embeddings

$$x_i^{l+1} = x_i^l + \phi_x(g_0 \oplus g^{l+1} \oplus \frac{c}{N} \sum_{i,j \in [N]} \phi_{ex}(m_{ij}) \oplus \sum_{i,j \in [N]} \phi_{ex}(m_{ij}))$$

5.5 Pass the results g^{l+1} and x_i^{l+1} to the next layer

6. After the final LE layer, run the particle representations through a final round of message-passing with the final global embedding

$$x_i^{L+1} = x_i^L + \phi_x(g_0 \oplus g^L \oplus \frac{c}{N} \sum_{i,j \in [N]} \phi_{ex}(m_{ij}) \oplus \sum_{i,j \in [N]} \phi_{ex}(m_{ij}))$$

References I

- [1] Shiqi Gong et al. “An Efficient Lorentz Equivariant Graph Neural Network for Jet Tagging”. In: *Journal of High Energy Physics* 2022.7 (July 2022), p. 30. ISSN: 1029-8479. DOI: 10.1007/JHEP07(2022)030. (Visited on 06/30/2025).
- [2] Lemeng Wu et al. “Fast Point Cloud Generation with Straight Flows”. In: *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Vancouver, BC, Canada: IEEE, June 2023, pp. 9445–9454. ISBN: 979-8-3503-0129-8. DOI: 10.1109/CVPR52729.2023.00911. (Visited on 06/23/2025).